

M57-JEAN DIGITAL FORENSICS AUTOPSY INVESTIGATION

Portfolio Case Study + SOC / DFIR Interview Playbook

Case	Platform	Evidence	Scope
M57-Jean Investigation	Autopsy 4.23.1 / The Sleuth Kit	nps-2008-jean.E01 training image	Labs 1-7: setup, ingestion, NTFS, keyword/hash, browser, email, timeline

Purpose: A consolidated study and portfolio document built from the seven M57-Jean lab modules and the hands-on investigation completed in Autopsy. It separates lab-supported procedures from analyst interpretation and records interface limitations encountered during the exercise.

Prepared for entry-level SOC Analyst / Digital Forensics interview preparation

How to Use This Guide

- For interviews: study Sections 2, 3, 10, 11, 12 and the rapid-fire Q&A.
- For GitHub/portfolio: use the case narrative, methodology, evidence table, limitations, and resume bullets.
- For future investigations: follow the triage decision flow rather than searching artifacts randomly.
- For revision: use the glossary and file-format reference as a quick cheat sheet.

Evidence discipline: A forensic artifact is not automatically a conclusion. Record what the artifact shows, correlate it with independent sources, then state the supported inference and its limitations.

1. Executive Case Summary

This project used the M57-Jean forensic training image to practice an end-to-end workstation investigation in Autopsy. The workflow progressed from forensic environment setup and evidence ingestion through NTFS/file-system analysis, keyword and hash-based triage, browser artifacts, email communications, and timeline correlation.

The strongest hands-on correlation developed during the later labs was a July 19 communication sequence involving Jean, Alison, and an external Gmail address, followed by system/device activity visible in the timeline. The exercise demonstrated why a defensible investigation is built from multiple artifacts rather than a single email, file, or timestamp.

Area	What was practiced	Why it matters
Evidence integrity	Disk image workflow, hashes, read-only forensic mindset	Protects authenticity and repeatability
File system	NTFS, MFT concepts, MAC times, deleted/unallocated data, signatures	Establishes what existed and what changed
Triage/search	Keywords, regex, hashes, known-good/known-bad concepts	Reduces noise and prioritizes leads
Web	History, downloads, searches, cookies, browser databases	Reconstructs user intent and online actions
Email	PST/MBOX/EML, headers, attachments, deleted mail, communication matrix	Builds communication and transfer context
Timeline	Cross-source chronological correlation	Turns artifacts into an event sequence

2. Investigator Triage Flow - What I Look At First

Triage is the rapid, hypothesis-driven process of deciding what deserves deeper examination. The goal is not to prove a theory immediately; it is to identify high-value leads, preserve context, and choose the next artifact that can confirm or challenge the current hypothesis.

Step	Question	First checks	Decision
1. Scope	What incident am I investigating?	Alert/case statement, user, host, suspected date range, business impact	Define hypothesis and boundaries
2. Time	When did the suspicious activity occur?	Timezone, alert time, email Date/Received, file MAC times, browser timestamps	Create a narrow investigation window
3. Identity	Who/what account or host is involved?	Username, email addresses, host/device identifiers	Pivot on account and machine
4. Network	Is there a suspicious external source/destination?	IP, domain, URL, email Received chain, download source	Check reputation/context; pivot to related artifacts
5. User action	What did the user do around	Web history/search, email,	Build behavioral sequence

	that time?	recent files, program execution, downloads	
6. File/object	What file was created, opened, sent, downloaded, deleted, or attached?	Path, type/signature, hash, timestamps, metadata	Determine relevance and integrity
7. Persistence/deletion	Was evidence removed or hidden?	Deleted files/email, unallocated space, extension mismatch	Recover/corroborate
8. Correlation	Do independent sources agree?	Email + file + browser + device + timeline	Raise or lower confidence
9. Document	Can another analyst reproduce this?	Exact path, timestamp/timezone, artifact source, screenshots/export, notes	Produce defensible finding

SOC-style triage shortcut: If an alert gives an IP and timestamp, start with those pivots. If it gives a user and file, pivot on user + filename/hash. If it gives an email, start with From/To/Subject/Date, then Received headers, attachment, URL/domain, and nearby endpoint activity.

Triage Decision Logic

- Benign/expected: artifact fits known user behavior and has corroborating legitimate context -> document and deprioritize.
- Suspicious but unconfirmed: anomaly exists but attribution/action is unclear -> collect more independent artifacts.
- High confidence: multiple independent sources support the same sequence -> escalate/report with evidence and limitations.
- Inconclusive: evidence is missing, overwritten, unavailable, or tool-limited -> say exactly that; do not fill the gap with assumptions.

3. End-to-End Forensic Analysis Flow

1. Preserve and verify evidence: work from a forensic image; validate hashes where available; record timezone and image/source identifiers.
2. Create/open case: establish case metadata and examiner notes; add the image as a data source.
3. Run ingest: select modules relevant to the hypothesis; wait for completion; note any failed/disabled modules.
4. Baseline the file system: identify OS/user profile, partitions, important directories, allocated/deleted/unallocated content.
5. Run targeted triage: keywords, regex, hashes, interesting files, extension mismatches.
6. Reconstruct user activity: browser history/downloads/searches/cookies, email messages/attachments, recent activity.
7. Correlate time: narrow to the incident window and compare events from independent sources.
8. Validate suspicious objects: file signature vs extension, hash, metadata, path, source, surrounding activity.
9. Record limitations: unavailable exports, missing artifacts, unsupported views, incomplete advanced steps.
10. Report: facts first, then inference, confidence, and next investigative action.

4. Labs 1-7 - Consolidated Technical Digest

Lab 1 - Introduction & Installation

Autopsy architecture, The Sleuth Kit, disk images, forensic phases, write blocking, hash verification, interface orientation.

Interview takeaway: Key interview point: Autopsy is the GUI/case manager; TSK performs core file-system parsing. Evidence is analyzed from images/copies, not altered originals.

Lab 2 - Case Management & Evidence Ingestion

Create case, add E01, set timezone, configure ingest modules, monitor ingest, understand autopsy.db/ModuleOutput/Reports.

Interview takeaway: Key interview point: ingest modules automate extraction and findings persist in the SQLite case database.

Lab 3 - File System Analysis

NTFS/MFT, MAC times, metadata, deleted files, unallocated space, carving, file signatures and extension mismatches.

Interview takeaway: Key interview point: deletion usually marks metadata/clusters reusable; recoverability depends on overwrite.

Lab 4 - Keyword Search & Ingest Modules

Custom keyword lists, regex, bookmarks, NSRL known-good, custom known-bad hashes, EXIF concepts.

Interview takeaway: Key interview point: triage combines content search and hash intelligence to reduce noise and surface high-value objects.

Lab 5 - Web & Browser Artifact Analysis

Web history, downloads, searches, cookies, Chrome/Firefox/IE artifact stores, SQLite browser databases.

Interview takeaway: Key interview point: browser artifacts can establish intent, source URL, destination path and timing.

Lab 6 - Email & Communication Forensics

PST/MBOX/EML, message review, attachments, deleted email, headers, sender IP concept, communication matrix, webmail cache search.

Interview takeaway: Key interview point: email is analyzed as content + headers + attachment + surrounding endpoint activity.

Lab 7 - Timeline Analysis

Event density, filtering, List view, chronological correlation, CSV/TimelineExplorer concept, \$SI vs \$FN timestomp concept.

Interview takeaway: Key interview point: timelines are only as reliable as their sources; corroborate timestamps and timezone.

5. Case Findings and Hands-On Evidence Notes

The following reflects the artifacts observed during the hands-on work, not a claim beyond what the training evidence supported.

Observed item	Timestamp / context	Analyst significance
Jean -> Alex: 'RE: which email address are you using?'	2008-07-19 18:44 CDT	Shows Jean-Alex communication and account/address discussion.
Jean -> Alison: 'This is what I was talking about'	2008-07-06 02:55 CDT	Very early-morning communication; useful as an after-hours pattern lead.
Jean/Alison thread follow-ups	2008-07-07 around 00:24-00:25 CDT	Additional after-hours communications.
Alison external Gmail -> Jean: request for information	2008-07-19 20:22:45 CDT	External personal-email address observed immediately before a response/attachment chain.
Jean -> Alison: 'RE: Please send me the information now'	2008-07-19 20:28 CDT	Message recorded with attachment present; m57biz.xls noted in the communication matrix.
Timeline: external request email	2008-07-19 20:22:45	Email Sent/Received artifacts visible in List view.
Timeline: device attached	2008-07-19 20:26:18	iCreate Technologies Corp. Flash Disk 256 MB observed; temporal proximity makes it a correlation lead, not by itself proof of transfer.
Google Alerts / external automated mail	2008-07-16 13:55:25 CDT	Useful comparison example of routine external automated communication.

Strong forensic wording: The timeline shows temporal proximity between an external information request, a removable-device attachment event, and Jean's subsequent email response. This supports further examination of the attachment, removable-media artifacts, and file access around the same window. Temporal proximity alone does not prove that a specific file was copied to the device.

Lab 7 Limitations Recorded

- Timeline CSV export: the lab PDF described File -> Export CSV, but that control was not exposed in the Autopsy 4.23.1 Timeline interface used during the exercise. The analysis itself was completed in Counts/List views; the CSV/TimelineExplorer checkpoint was not represented as completed.
- Advanced timestamp comparison: the required NTFS \$MFT was not directly exposed through the attempted view/search; searching '\$MFT' returned carved/content hits rather than the metadata file needed for MFTECmd. Therefore \$STANDARD_INFORMATION vs \$FILE_NAME comparison for five files was not represented as completed.

Why this helps in an interview: A strong analyst does not manufacture a result to satisfy a checklist. Tool/version limitations are documented, and the analyst identifies the exact evidence needed to complete the step.

6. Special Forensic File-System and Artifact Structure

Artifact / format	Full form / structure	Primary use	Forensic value
.E01	EnCase Expert Witness Format	Forensic disk image container; may be segmented .E01/.E02/...	Preserves a bitstream image plus metadata/checksums; Autopsy/TSK can parse it.
NTFS	New Technology File System	Windows file system	MFT records, timestamps, allocation state, metadata and deleted-file analysis.
\$MFT	Master File Table	NTFS metadata database	At least one record per

			file/directory; names, attributes, timestamps, data references.
\$STANDARD_INFORMATION (\$SI)	NTFS attribute	Standard metadata/timestamps	Common timestamp source; can be modified via normal APIs/tools.
\$FILE_NAME (\$FN)	NTFS attribute	Filename/parent reference and timestamps	Useful comparison source because timestamps are harder to manipulate through normal APIs.
Unallocated space	Clusters not currently assigned to active files	Free disk area	May retain deleted content until overwritten.
Slack space	Unused bytes within an allocated cluster	Residual area	Can contain remnants of prior data.
.PST	Personal Storage Table	Microsoft Outlook mail store	Messages, folders, contacts, calendar, attachments; deleted content may persist.
MBOX	Mailbox format	Flat text mail archive	Messages concatenated in one file; highly searchable and sometimes recoverable in remnants.
.EML	Electronic Mail message file	Standalone RFC-style email	Headers, body, MIME parts and attachments.
MIME	Multipurpose Internet Mail Extensions	Defines email content types/encodings and multipart boundaries	Explains how HTML, text and attachments are represented inside email.
.CSV	Comma-Separated Values	Tabular text export	Portable evidence matrix/timeline export; easy to filter but does not itself preserve forensic provenance.
SQLite / .sqlite / .db	SQLite database	Browser/application structured data	History, cookies, form data and Autopsy case data may be stored in SQLite.
places.sqlite	Firefox SQLite database	History/bookmarks	moz_places, moz_historyvisits and bookmark relationships.
cookies.sqlite	Firefox cookie database	Cookie records	Domain, cookie names/values, expiry and session context.
History	Chromium SQLite database	URLs, visits, downloads	Reconstructs browsing and download activity.
WebCacheV01.dat	ESE database used by legacy Microsoft web components	IE/legacy Edge web cache/history	Can hold browser activity artifacts.
index.dat	Legacy Internet Explorer artifact	Older IE history/cache records	Historical web-activity source on older systems.
Prefetch (.pf)	Windows Prefetch file	Program execution optimization artifact	Can support evidence that an executable ran and provide execution-related timing/context.
EXIF	Exchangeable Image File Format metadata	Image metadata	Camera/software/date and sometimes GPS information.
MD5 / SHA-256	Message Digest 5 / Secure Hash Algorithm 256-bit	Cryptographic hash/checksum	Integrity verification, deduplication, known-good/known-bad matching.
HTML	HyperText Markup Language	Web/email rendered content	Carved cache pages can reveal prior webmail/web activity.
Registry hive	Windows Registry database files	OS/user configuration	Last-write times and configuration artifacts can corroborate endpoint activity.

7. Email Forensics Deep-Dive

Email triage order

11. From / Reply-To / Return-Path
12. To / CC / BCC where available
13. Subject and conversation/thread context
14. Date plus Received: chain (read relay path carefully)
15. Message-ID and other identifiers
16. URLs/domains and displayed-vs-actual links
17. Attachments: filename, MIME type, signature, hash, metadata
18. Body language and request/action
19. Deleted-folder state or webmail/cache remnants
20. Endpoint activity around the same timestamp

Received header principle: The lab emphasizes using the Received chain to understand relay path and looking for X-Originating-IP when present. Do not treat the visible From address alone as proof of origin.

MIME in plain English

MIME lets one email carry multiple content parts: plain text, HTML, inline images and attachments. A multipart message uses boundaries to separate parts. Content-Type describes what a part claims to be; Content-Transfer-Encoding describes how it is encoded (for example, base64). Forensic analysis compares the declared type/filename with actual file signature and context.

8. Browser Forensics Deep-Dive

Question	Artifact to check	What it can establish
What sites were visited?	Web History / History / places.sqlite	URL, title, visit timing and browser source
What was downloaded?	Web Downloads / browser History	Source URL, destination, times, size; compare with file system
What did the user search?	Web Search / query URLs	Potential intent or research context
Was the user logged in?	Cookies/session records	Domain/session context; not automatic proof of who physically used the machine
Was webmail used?	History/cache/carved HTML	Can reveal Gmail/webmail access or cached content
Does the downloaded file still exist?	Destination path + file system	If absent, investigate deletion/unallocated space rather than assuming why

9. File-System and Timestamp Analysis

Timestamp	Meaning	Use with caution
Modified (M)	File content last changed	Can be changed/manipulated; correlate.
Accessed (A)	File last opened/read	OS settings may reduce/update behavior; do not over-interpret.
Changed (C / MFT changed)	Metadata/MFT record changed	Can reveal metadata operations.
Created (B/Born)	Created on current volume	Copying can reset creation time; moving within volume behaves differently.

For advanced timestamp analysis, the lab compares \$STANDARD_INFORMATION and \$FILE_NAME Created times. A discrepancy can be a lead, but the analyst should document both values and consider legitimate explanations before concluding manipulation.

Timestamp rule: Never build a critical conclusion from one timestamp source. Confirm timezone and correlate file-system times with email, browser, registry, event logs, prefetch, USN Journal or other independent artifacts when available.

10. Strong SOC / DFIR Interview Answers

Walk me through your forensic triage process.

I start by defining the incident scope, user/host, timezone and the narrowest reliable time window. Then I pivot on the strongest indicators - for example an IP, domain, account, filename or hash. I examine high-value artifacts around that window, correlate independent sources such as email, browser and file-system activity, validate suspicious files by signature/hash/metadata, and document facts separately from inference. If evidence is missing or a tool cannot expose an artifact, I record the limitation instead of forcing a conclusion.

What did you learn from the M57-Jean project?

I learned to move from artifact collection to correlation. The strongest part was not finding a single email; it was lining up communication timestamps with endpoint activity in Autopsy's timeline. I also learned that tool versions can change the workflow, so I document what was actually available and keep the result reproducible.

How would you investigate a suspicious email?

I validate sender and recipient fields, inspect the Received chain and Message-ID, look for Reply-To or origin anomalies, extract URLs/domains, inspect attachments by file signature and hash, then correlate the message time with endpoint activity such as downloads, file access, process execution and removable media. I do not rely only on the visible From field.

How do you handle deleted files?

I first distinguish metadata-based deleted entries from carved content. In NTFS, deletion marks the MFT entry and clusters reusable; data may survive until overwritten. I inspect metadata and content, recover when possible, hash the recovered object, and correlate its path/timestamps with other artifacts.

What is the difference between extension and file type?

An extension is a filename label. The actual type is identified from content structure or magic bytes. A renamed executable can still be detected as an executable, which is why extension mismatch analysis matters.

What is the purpose of a hash?

For forensics, hashes support integrity verification and file identification. I can compare evidence hashes across stages and use hash sets to classify known-good or known-bad files. A hash match identifies identical content for that algorithm; it does not by itself explain how or why the file got there.

How do you build a timeline?

I normalize the timezone, identify the incident window, then combine events from multiple sources - file MAC times, browser activity, email, registry/event logs and execution artifacts where available. I use event density to find spikes, filter by type, and then inspect the individual events in chronological order.

What would make you escalate a finding?

I look for corroboration and impact. One anomalous artifact is a lead; multiple independent artifacts that support the same suspicious sequence, especially involving sensitive data, external communication, execution or removable media, increase confidence and justify escalation.

11. Interview Story - STAR Format

STAR	Answer
Situation	I built a hands-on Autopsy case using the M57-Jean training image to practice workstation digital forensics from ingestion through timeline analysis.
Task	My goal was to identify and correlate relevant file-system, browser and email artifacts and produce a defensible event narrative rather than just collect screenshots.
Action	I configured ingest modules, reviewed NTFS/deleted data, used targeted keyword searches, analyzed browser artifacts, exported and reviewed email records, built a communication matrix, and narrowed the Timeline List view to the July 19 activity window. I correlated an external information request with a removable-device attachment event and a later email response, while documenting that temporal proximity alone did not prove a file copy.
Result	I produced a repeatable investigation workflow, evidence notes and a portfolio-ready case narrative. I also documented two advanced Lab 7 limitations instead of overstating completion: Timeline CSV export was unavailable in my interface and the required \$MFT was not directly exposed for the \$SI/\$FN comparison.

12. Recruiter and Interview Opening Scripts

30-second recruiter intro: I'm transitioning into cybersecurity with a primary focus on SOC Analyst L1 and a secondary interest in digital forensics. I hold the Google Cybersecurity Professional Certificate and ISC2 CC, and I've been building hands-on projects in Autopsy, Kali Linux and VirtualBox. My recent Autopsy project covered evidence ingestion, NTFS and deleted-file analysis, browser and email artifacts, and timeline correlation. I'm looking for an entry-level role where I can apply that investigation mindset and continue developing in a real SOC environment.

Technical interviewer opening: One project I'd be happy to walk through is my M57-Jean Autopsy investigation. I treated it as an end-to-end forensic case: I verified the evidence workflow, configured ingest, examined NTFS and deleted artifacts, performed keyword and browser analysis, reviewed PST email communications and attachments, and then correlated the most relevant activity in the timeline. I can walk you through both the findings and the limitations I documented.

13. Must-Say Phrases for Interviews

- I separate observed facts from analyst inference.
- I normalize and verify timezone before correlating timestamps.
- I pivot from the alert's strongest indicator - user, host, IP, domain, file, hash or timestamp.
- I corroborate important findings with independent artifact sources.
- I validate file type using content/signature, not only the extension.
- I preserve evidence integrity and work from forensic copies/images.

- I document tool limitations and incomplete evidence rather than guessing.
- A suspicious artifact is a lead; context and correlation determine its significance.
- Temporal proximity supports correlation, but it does not by itself prove causation.
- I record exact paths, timestamps, artifact source and reproducible steps.

14. Commands and Technical Cheat Sheet

Command / query	Use
MFTECmd.exe -f <path-to-\$MFT> --csv <output> --csvf MFT_analysis.csv	Parse NTFS MFT to CSV for attribute/timestamp comparison (advanced lab concept).
Import-Csv MFT_analysis.csv Where-Object { \$_.'SI Created' -lt \$_.'FN Created' }	Lab example to surface SI/FN Created discrepancies.
readpst -o <output_dir> <file.pst>	Extract PST contents to a directory/MBOX-style output on Linux.
readpst -D <file.pst>	List PST folders (lab workflow).
readpst -f "Deleted Items" <file.pst>	Target Deleted Items extraction (lab workflow).
fls -r -m C: <image> > bodyfile.txt	TSK-style recursive body-file generation concept for timeline work.
mactime -b bodyfile.txt -d > timeline.csv	Convert body file to CSV timeline.
SELECT url, title, visit_count, last_visit_time FROM urls ORDER BY last_visit_time DESC LIMIT 50;	Lab example for querying Chromium History database.
Regex email: \b[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}\b	Find email-address patterns.
Regex IPv4: \b(?:\d{1,3}\.){3}\d{1,3}\b	Find IPv4-looking patterns; validate octet ranges separately.

15. Glossary - Full Forms and Uses

Term	Full form	Use / meaning
DFIR	Digital Forensics and Incident Response	Investigation and response discipline combining evidence analysis with containment/recovery context.
SOC	Security Operations Center	Team/function that monitors, triages and responds to security events.
TSK	The Sleuth Kit	Open-source forensic engine used by Autopsy for file-system analysis.
NTFS	New Technology File System	Windows file system with MFT-based metadata.
MFT	Master File Table	Core NTFS metadata database.
MAC times	Modified, Accessed, Changed, Created/Born timestamps	File-system timing evidence; naming conventions vary.
SI	\$STANDARD_INFORMATION	NTFS attribute containing standard metadata/timestamps.
FN	\$FILE_NAME	NTFS attribute containing filename/parent reference and timestamps.
USN	Update Sequence Number Journal	NTFS change journal useful for file-system activity reconstruction.
PST	Personal Storage Table	Outlook data container.
MBOX	Mailbox	Flat-file email storage format used by several mail clients.
EML	Email message file	Standalone email message format.
MIME	Multipurpose Internet Mail Extensions	Email content/attachment typing and encoding framework.
SMTP	Simple Mail Transfer Protocol	Protocol used to send/relay email.
POP3	Post Office Protocol version 3	Mail retrieval protocol, commonly downloads mail to a client.
IMAP	Internet Message Access Protocol	Mail retrieval/synchronization protocol retaining server-side folder state.
EXIF	Exchangeable Image File Format	Image metadata standard.

NSRL	National Software Reference Library	NIST reference hashes for known software files.
RDS	Reference Data Set / Reduced Data Set in NSRL context	Reference hash dataset used to classify known files.
MD5	Message Digest Algorithm 5	128-bit hash, widely used in forensics for identification/integrity though not collision-resistant for security design.
SHA-256	Secure Hash Algorithm 256-bit	Cryptographic hash used for stronger integrity verification.
CSV	Comma-Separated Values	Portable tabular text format.
SQLite	Lightweight embedded relational database	Common browser/app storage and Autopsy case database technology.
ESE	Extensible Storage Engine	Microsoft database engine used by artifacts such as WebCacheV01.dat.
URL	Uniform Resource Locator	Web resource address.
IP	Internet Protocol address	Network endpoint identifier.
UTC	Coordinated Universal Time	Time standard useful for normalizing forensic events.
IOC	Indicator of Compromise	Observable value such as hash/domain/IP/path that may be associated with malicious activity.
TTP	Tactics, Techniques, and Procedures	Description of adversary behavior patterns.

16. Portfolio / GitHub Case-Study Template

Suggested repository structure:

- README.md - case objective, tools, methodology, key findings, limitations
- screenshots/ - only purposeful evidence screenshots with neutral filenames
- notes/ - investigation notes and artifact tables
- exports/ - sanitized CSVs or non-sensitive training outputs where appropriate
- docs/ - final case-study PDF

README summary example

Analyzed the M57-Jean forensic training image in Autopsy 4.23.1 using an end-to-end workflow covering evidence ingestion, NTFS/file-system artifacts, deleted data, targeted keyword searches, browser artifacts, email communications and timeline correlation. Built a communication matrix and correlated July 19 email activity with endpoint/removable-device events. Documented tool/version limitations for advanced timeline export and MFT timestamp comparison rather than overstating results.

17. Resume Bullets

- Performed end-to-end digital forensic analysis of the M57-Jean training disk image in Autopsy, covering evidence ingestion, NTFS metadata, deleted files, browser artifacts, email/PST analysis, and timeline reconstruction.
- Built a 10-record communication matrix and correlated email, attachment, timestamp, and endpoint activity to reconstruct a high-interest July 19 event sequence.
- Applied keyword/regex triage, file-signature validation, hashing concepts, and cross-artifact correlation while documenting evidence provenance, timezone, and tool limitations.
- Analyzed web history, downloads, searches, cookies, email headers, and removable-device timeline events to develop reproducible investigative leads.

18. Rapid-Fire Interview Q&A

Question	Strong concise answer
Allocated vs unallocated?	Allocated clusters belong to active files; unallocated clusters are not currently assigned and may retain deleted remnants.
Carving?	Recovering data based on file signatures/structure when normal file-system metadata is missing or unusable.
Why magic bytes?	They identify the actual content type and can expose misleading extensions.
Why NSRL?	To identify known software files and reduce analyst noise.
Known-good vs known-bad?	Known-good can be deprioritized; known-bad/notable hashes flag files of interest.
Why Message-ID?	It is a message identifier that can help correlate copies across mail systems/artifacts.
Why read Received headers?	They record mail relay hops and provide stronger routing context than the visible From field alone.
Why timeline correlation?	It places independent artifacts in chronological context to reconstruct behavior.
Why is timezone important?	A wrong timezone can make unrelated events appear correlated or hide the true sequence.
What is timestomping?	Intentional modification of file timestamps to mislead analysis.
What is evidence provenance?	Knowing exactly which source/image, path, artifact parser and extraction produced a finding.
What if a tool cannot export?	Document the version/interface limitation, preserve the analysis you can reproduce, and use an equivalent method only if it truly produces the same evidence type.

19. Final Investigation Checklist

- Case scope/user/host recorded
- Evidence image/source and integrity information recorded
- Timezone confirmed
- Ingest status/modules recorded
- High-value keywords/IOCs searched
- Relevant files validated by path/type/hash/metadata
- Deleted/unallocated evidence considered
- Browser history/download/search/cookie artifacts reviewed
- Email headers/body/attachments/deleted items reviewed
- Timeline narrowed to incident window
- At least two independent sources used for major findings
- Facts separated from inference
- Limitations documented
- Evidence screenshots/exports named and organized
- Case narrative can be reproduced by another analyst

20. Source Basis and Scope Notes

The lab procedures and terminology in this guide are grounded in the seven M57-Jean Notion lab PDFs used for the project. Additional explanatory sections - such as the investigator triage decision flow, interview answers, portfolio wording, expanded glossary, and practical cautions - are synthesis added for study and career preparation. Where the hands-on Autopsy 4.23.1 interface differed from the lab PDF, the actual observed limitation is preserved rather than silently treating the PDF workflow as completed.